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COSI — Schemas and Samples

VDW
Die Wellpappenindustrie



COSI 1.0
Corrugated Standard Interface

Open industry standard for machine data exchange
in corrugated cardboard manufacturing

Imprint

COSI — Schemas and Samples

COSI — Corrugated Standard Interface

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VDW (Association of the German Corrugated Cardboard Industry e.V.)

Authors

Michael Sachs

Robert Schmid

Daniel Malolepszy

Dennis Strasser

Tobias Winkler

Technical editors

Technical specification development and quality assurance.

André Seifert (ePS)

Robin Boeckxstaens (OMP)

Contact

Christiane Trumpp

transparente-daten@vdw-da.de

Disclaimer

The PDF edition is a reproduction of the JSON schemas and samples published in the official COSI GitHub repository. In the event of any discrepancy, the JSON source for the corresponding document version shall prevail.

https://github.com/vdw-da/COSI-Corrugated-Standard-Interface

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1 Schemas

1.1 common-types.schema.json

Source: schemas/common-types.schema.json

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "cosi:common-defs:1.0",
  "title": "COSI -- Common Definitions",
  "description": "Shared type definitions referenced by other payload schemas.",

  "$defs": {

    "Quantity": {
      "description": "A measured value together with its unit of measure.",
      "type": "object",
      "required": ["value", "uom"],
      "properties": {
        "value": { "type": "number" },
        "uom":   { "type": "string" }
      },
      "additionalProperties": false
    },

    "CustomProps": {
      "description": "Manufacturer-/site-specific extension properties. Content is not
      ↪  standardized by this specification; consumers must ignore unknown keys.",
      "type": "object"
    },

    "QuantityMM": {
      "description": "A measurement in millimetres.",
      "type": "object",
      "required": ["value", "uom"],
      "properties": {
        "value": { "type": "number" },
        "uom":   { "type": "string", "const": "mm" }
      },
      "additionalProperties": false
    }

  }
}
```

1.2 envelope.schema.json

Source: schemas/envelope.schema.json

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "cosi:envelope:1.0",
  "title": "COSI -- Message Envelope",
  "description": "Top-level wrapper for all COSI messages. Common fields are validated here; the payload is validated against the schema that matches assetClass.",
  "type": "object",
  "required": [
    "messageID",
    "assetClass",
    "schemaId",
    "messageType",
    "machineId",
    "timestamp",
    "status",
    "payload"
  ],
  "additionalProperties": false,
  "properties": {
    "messageID": {
      "type": "string",
      "format": "uuid",
      "pattern": "^[0-9a-f]{8}-[0-9a-f]{4}-5[0-9a-f]{3}-[89ab][0-9a-f]{3}-[0-9a-f]{12}$",
      "description": "Globally unique UUID v5 for this message instance. Producers allocate distinct generation inputs for distinct instances, even with identical content. Retries and replays preserve the assigned ID, including across restart; UUID v5 alone does not ensure uniqueness or de-duplication."
    },
    "correlationID": {
      "type": "string",
      "format": "uuid",
      "pattern": "^[0-9a-f]{8}-[0-9a-f]{4}-5[0-9a-f]{3}-[89ab][0-9a-f]{3}-[0-9a-f]{12}$",
      "description": "UUID v5 linking related messages. Required on setup-response: the messageID of the original setup request."
    },
    "assetClass": {
      "type": "string",
      "enum": [
        "performance",
        "setup-corrugator",
        "setup-converting",
        "setup-response",
        "maintenance-document",
        "maintenance-measurement",
        "quality",
        "lastwill"
      ],
      "description": "Identifies the business domain of this message. Determines which payload schema applies."
    },
    "messageType": {
      "type": "string",
      "enum": [
        "hbt",
        "cos",
        "onetime"
      ],
      "description": "hbt = Heartbeat (periodic, min every 5 s). cos = Change-of-Status (event-driven). onetime = Static reference data sent once."
    },
    "machineId": {
      "type": "string",
      "minLength": 1,
      "description": "Plant-owned machine identifier, unique within its site and matching the machine topic segment. Providers map their identifiers to the plant definition.",
      "pattern": "^[a-z0-9-]+$"
    },
    "timestamp": {
```

```
      "oneOf": [
        {
          "type": "string",
          "format": "date-time",
          "pattern": "^\\d{4}-\\d{2}-\\d{2}T\\d{2}:\\d{2}:\\d{2}(\\.\\d{1,3})?Z$",
          "description": "ISO 8601 UTC timestamp of the event being reported (not the time
          ↪  the message was created or transmitted). The Z suffix is mandatory; local-time
          ↪  offsets are not allowed (e.g., 2026-04-01T08:23:11.000Z)."
        },
        {
          "type": "null"
        }
      ],
      "description": "Original event time in UTC Z form; required null only for the dedicated
      ↪  Last Will, whose future publication time is unknown at registration."
    },
    "status": {
      "type": "string",
      "enum": [
        "unknown",
        "stop",
        "run",
        "fault",
        "setup",
        "maintenance",
        "offline",
        "lastwill"
      ],
      "description": "Machine state at the event time. offline is the final
      ↪  controlled-shutdown notification; lastwill is reserved for the dedicated
      ↪  connection-loss notification and does not assert a physical stop."
    },
    "payload": {
      "type": "object",
      "description": "Asset-class-specific data. Schema is determined by assetClass (enforced
      ↪  by the oneOf below)."
    },
    "schemaId": {
      "type": "string",
      "description": "Full payload schema identifier, including version; must match
      ↪  assetClass."
    },
    "connectionId": {
      "type": "string",
      "format": "uuid",
      "pattern": "^[0-9a-f]{8}-[0-9a-f]{4}-5[0-9a-f]{3}-[89ab][0-9a-f]{3}-[0-9a-f]{12}$",
      "description": "Required on the Will and normal equipment messages of a connection
      ↪  using Last Will. Fresh globally unique UUID v5 per MQTT connection, including
      ↪  reconnect/restart; shared for that connection. Distinct from MQTT client ID and
      ↪  messageID. Allocation method is implementation-defined."
    }
  },
  "oneOf": [
    {
      "properties": {
        "assetClass": {
          "const": "performance"
        },
        "payload": {
          "$ref": "cosi:performance-payload:1.0"
        },
        "schemaId": {
          "const": "cosi:performance-payload:1.0"
        }
      }
    },
    {
      "properties": {
        "assetClass": {
          "const": "setup-corrugator"
        },
        "payload": {
          "$ref": "cosi:setup-corrugator-payload:1.0"
        },
```

```
        "schemaId": {
          "const": "cosi:setup-corrugator-payload:1.0"
        }
      }
    },
    {
      "properties": {
        "assetClass": {
          "const": "setup-converting"
        },
        "payload": {
          "$ref": "cosi:setup-converting-payload:1.0"
        },
        "schemaId": {
          "const": "cosi:setup-converting-payload:1.0"
        }
      }
    },
    {
      "properties": {
        "assetClass": {
          "const": "setup-response"
        },
        "payload": {
          "$ref": "cosi:setup-response-payload:1.0"
        },
        "schemaId": {
          "const": "cosi:setup-response-payload:1.0"
        }
      },
      "required": [
        "correlationID"
      ]
    },
    {
      "properties": {
        "assetClass": {
          "const": "maintenance-document"
        },
        "payload": {
          "$ref": "cosi:maintenance-document-payload:1.0"
        },
        "schemaId": {
          "const": "cosi:maintenance-document-payload:1.0"
        }
      }
    },
    {
      "properties": {
        "assetClass": {
          "const": "maintenance-measurement"
        },
        "payload": {
          "$ref": "cosi:maintenance-measurement-payload:1.0"
        },
        "schemaId": {
          "const": "cosi:maintenance-measurement-payload:1.0"
        }
      }
    },
    {
      "properties": {
        "assetClass": {
          "const": "quality"
        },
        "payload": {
          "$ref": "cosi:quality-payload:1.0"
        },
        "schemaId": {
          "const": "cosi:quality-payload:1.0"
        }
      }
    },
    {
```

```
      "properties": {
        "assetClass": {
          "const": "lastwill"
        },
        "schemaId": {
          "const": "cosi:lastwill-payload:1.0"
        },
        "payload": {
          "$ref": "cosi:lastwill-payload:1.0"
        }
      }
    }
  ],
  "allOf": [
    {
      "if": {
        "properties": {
          "assetClass": {
            "const": "lastwill"
          }
        },
        "required": [
          "assetClass"
        ]
      },
      "then": {
        "required": [
          "connectionId"
        ],
        "properties": {
          "timestamp": {
            "type": "null"
          },
          "status": {
            "const": "lastwill"
          },
          "messageType": {
            "const": "cos"
          }
        }
      },
      "else": {
        "properties": {
          "timestamp": {
            "type": "string",
            "format": "date-time",
            "pattern": "^\\d{4}-\\d{2}-\\d{2}T\\d{2}:\\d{2}:\\d{2}(\\.\\d{1,3})?Z$",
            "description": "ISO 8601 UTC timestamp of the event being reported (not the time
            ↪  the message was created or transmitted). The Z suffix is mandatory;
            ↪  local-time offsets are not allowed (e.g., 2026-04-01T08:23:11.000Z)."
          },
          "status": {
            "enum": [
              "unknown",
              "stop",
              "run",
              "fault",
              "setup",
              "maintenance",
              "offline"
            ]
          }
        }
      }
    }
  ]
}
```

1.3 lastwill.schema.json

Source: schemas/lastwill.schema.json

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "cosi:lastwill-payload:1.0",
  "title": "COSI -- Last Will Payload",
  "description": "Empty payload for the optional OT-equipment Last Will. Notification data is carried by the envelope (ADR-000022).",
  "type": "object",
  "properties": {},
  "additionalProperties": false
}
```

1.4 maintenance-document.schema.json

Source: schemas/maintenance-document.schema.json

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "cosi:maintenance-document-payload:1.0",
  "title": "COSI -- Maintenance Document Payload",
  "description": "Payload for maintenance-document messages (assetClass=maintenance-document). Carries manual PDF links and recurring service task definitions for one machine component.",
  "type": "object",
  "required": ["componentId", "manualDocs", "serviceTasks"],

  "$defs": {

    "DocumentRef": {
      "description": "Pointer to a document in an external system or file store.",
      "type": "object",
      "required": ["uniqueRef", "documentLink"],
      "properties": {
        "uniqueRef":    { "type": "string", "minLength": 1, "description": "Stable unique identifier for this document across systems." },
        "documentLink": { "type": "string", "minLength": 1, "description": "URI or file-system path to a locally accessible document. The reference does not authorize bypassing the IT/OT boundary." }
      },
      "additionalProperties": false
    }

  },

  "properties": {

    "componentId": {
      "type": "string",
      "minLength": 1,
      "description": "Unique identifier of the machine component this document record belongs to."
    },

    "manualDocs": {
      "type": "array",
      "items": {
        "type": "object",
        "required": ["documentRef", "validFrom"],
        "properties": {
          "documentRef": { "$ref": "#/$defs/DocumentRef" },
          "validFrom": {
            "type": "string",
            "format": "date-time",
            "description": "ISO 8601 UTC date from which this document version is valid."
          }
        },
        "additionalProperties": false
      },
      "description": "Maintenance manual documents for this component, ordered by document type or relevance."
    },

    "serviceTasks": {
      "type": "array",
      "items": {
        "type": "object",
        "required": ["taskId", "frequency", "counterSource", "documentRef"],
        "properties": {
          "taskId": {
            "type": "string",
            "minLength": 1,
            "description": "Unique identifier for this service task."
          },
          "frequency": {
```

```
            "type": "object",
            "required": ["value", "uom"],
            "properties": {
              "value": { "type": "number", "exclusiveMinimum": 0, "description": "Numeric interval after which the task must be performed." },
              "uom": {
                "type": "string",
                "enum": ["h", "m²", "pcs", "cycles"],
                "description": "Counter unit: h = machine runtime hours; m² = board area produced; pcs = pieces produced; cycles = machine cycles."
              }
            },
            "additionalProperties": false
          },
          "counterSource": {
            "type": "string",
            "description": "Name of the machine counter used to track frequency."
          },
          "documentRef": { "$ref": "#/$defs/DocumentRef" }
        },
        "additionalProperties": false
      },
      "description": "Recurring service tasks for this component, each with a frequency trigger and reference to the task instruction document."
    }

  },

  "additionalProperties": false
}
```

1.5 maintenance-measurement.schema.json

Source: schemas/maintenance-measurement.schema.json

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "cosi:maintenance-measurement-payload:1.0",
  "title": "COSI -- Maintenance Measurement Payload",
  "description": "Payload for maintenance-measurement messages (assetClass=maintenance-measurement). Carries a measured value against upper/lower limits with fault classification and a required document-reference object whose fields may be null.",
  "type": "object",
  "required": ["componentId", "componentName", "measuredValue", "upperLimit", "lowerLimit", "failure", "documentRef"],

  "$defs": {

    "MeasuredValue": {
      "description": "A measurement with its unit of measure.",
      "type": "object",
      "required": ["value", "uom"],
      "properties": {
        "value": { "type": "number" },
        "uom":   { "type": "string", "minLength": 1 }
      },
      "additionalProperties": false
    },

    "LimitValue": {
      "description": "A tolerance limit. value and uom are null when the limit is not defined.",
      "type": "object",
      "required": ["value", "uom"],
      "properties": {
        "value": { "type": ["number", "null"] },
        "uom":   { "type": ["string", "null"] }
      },
      "additionalProperties": false
    }

  },

  "properties": {

    "componentId": {
      "type": "string",
      "minLength": 1,
      "description": "Unique identifier of the component being measured."
    },

    "componentName": {
      "type": "string",
      "description": "Human-readable component name including location (e.g., 'SF1 Pressure Roll -- Drive-Side Bearing')."
    },

    "measuredValue": {
      "$ref": "#/$defs/MeasuredValue",
      "description": "Value observed at the original envelope timestamp. This message reports an observation, not an open/update/clear fault lifecycle."
    },

    "upperLimit": {
      "$ref": "#/$defs/LimitValue",
      "description": "Upper tolerance bound. Null value when no upper limit is defined."
    },

    "lowerLimit": {
      "$ref": "#/$defs/LimitValue",
      "description": "Lower tolerance bound. Null value when no lower limit is defined."
    },
```

```
    "failure": {
      "type": "object",
      "required": ["faultCode", "descLocalLang", "descEN", "severity"],
      "properties": {
        "faultCode": {
          "type": ["string", "null"],
          "enum": [null, "AIR", "BRD", "CLW", "DOP", "ELP", "ELU", "ERO", "FOF", "FOV", "FTC",
                   "FTF", "FTI", "FTO", "FTR", "FTS", "HIO", "ILP", "ILU", "INL", "LCP",
                   "LOO", "NOI", "OHE", "OTH", "PCL", "PDE", "PLU", "POW", "PTF", "SER",
                   "SHH", "SLL", "SLP", "SPO", "STD", "STP", "UNK", "UST", "VIB", "VLO"],
          "description": "Failure mode code from docs/corrugated-faults.md (closed set for v1.0). Null when the measurement is within limits."
        },
        "internalCode": {
          "type": ["string", "null"],
          "description": "Optional machine-vendor-specific internal fault code."
        },
        "descLocalLang": {
          "type": ["string", "null"],
          "description": "Fault description in the local plant language."
        },
        "descEN": {
          "type": ["string", "null"],
          "description": "Fault description in English."
        },
        "severity": {
          "type": ["string", "null"],
          "enum": ["info", "warning", "critical", null],
          "description": "info = minor, no production loss. warning = operator attention required. critical = immediate action / auto-stop."
        }
      },
      "additionalProperties": false
    },

    "documentRef": {
      "type": "object",
      "required": ["uniqueRef", "documentLink"],
      "properties": {
        "uniqueRef":    { "type": ["string", "null"], "description": "Reference to the maintenance manual document. Null if none available." },
        "documentLink": { "type": ["string", "null"], "description": "URI or file-system path. Null if none available." }
      },
      "additionalProperties": false,
      "description": "Required document-reference object; its fields may be null when no link is available. Access remains subject to the IT/OT boundary rules."
    }

  },

  "additionalProperties": false
}
```

1.6 performance.schema.json

Source: schemas/performance.schema.json

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "cosi:performance-payload:1.0",
  "title": "COSI -- Performance Payload",
  "description": "Payload for Performance messages (assetClass=performance). Validated as the payload property of the message envelope.",
  "type": "object",
  "required": ["speed", "inputs", "outputs"],

  "$defs": {

    "Quantity": { "$ref": "cosi:common-defs:1.0#/$defs/Quantity" },

    "StreamItem": {
      "description": "One input or output stream. Quantities are cumulative for the identified scope and counting period, regardless of hbt/cos trigger. Compare compatible readings; do not add successive totals. Article change alone does not reset a counter. See Technical Guide § 3.5.",
      "type": "object",
      "properties": {
        "name":          { "type": ["string", "null"] },
        "streamId":      { "type": ["string", "null"], "description": "Machine-internal stream identifier." },
        "jobRef":        { "type": ["string", "null"] },
        "articleRef":    { "type": ["string", "null"] },
        "area":          { "$ref": "#/$defs/Quantity", "description": "Area in m²." },
        "weight":        { "$ref": "#/$defs/Quantity", "description": "Weight in metric tonnes (t)." },
        "runningLength": { "$ref": "#/$defs/Quantity", "description": "Running length in m." },
        "pieces":        { "$ref": "#/$defs/Quantity", "description": "Cumulative discrete units at the defined counting point; use a consistent unit and counting basis within the period." },
        "stacks":        { "$ref": "#/$defs/Quantity", "description": "Cumulative count of physical stacks, not sheets per stack." }
      },
      "additionalProperties": false
    },

    "EnergyItem": {
      "description": "Cumulative resource counters for the identified machine, stream or job and counting period. Distinguish measured totals from allocated shares; do not add shares to their source total. Reporting measured totals alone is permitted.",
      "type": "object",
      "properties": {
        "name":        { "type": ["string", "null"] },
        "streamId":    { "type": ["string", "null"] },
        "jobRef":      { "type": ["string", "null"] },
        "articleRef":  { "type": ["string", "null"] },
        "water":       { "type": "object", "allOf": [{ "$ref": "#/$defs/Quantity" }], "properties": { "uom": { "const": "m³"     } } },
        "electricity": { "type": "object", "allOf": [{ "$ref": "#/$defs/Quantity" }], "properties": { "uom": { "const": "Wh"     } } },
        "steam":       { "type": "object", "allOf": [{ "$ref": "#/$defs/Quantity" }], "properties": { "uom": { "const": "kg"     } } },
        "air":         { "type": "object", "allOf": [{ "$ref": "#/$defs/Quantity" }], "properties": { "uom": { "const": "m³"     } } },
        "stacks":      { "type": "object", "allOf": [{ "$ref": "#/$defs/Quantity" }], "properties": { "uom": { "const": "stacks" } }, "description": "Cumulative physical stack count for the identified scope and period; not sheets per stack." }
      },
      "additionalProperties": false
    },

    "Speed": {
```

```
      "description": "Current and target machine speed. The target baseline is supplied by ERP/MES according to the plant's business rules; COSI does not prescribe its calculation method.",
      "type": "object",
      "required": ["current", "uom"],
      "properties": {
        "current": { "type": "number", "minimum": 0 },
        "target":  { "type": "number", "minimum": 0 },
        "uom": {
          "type": "string",
          "enum": ["pcs/h", "m/min"],
          "description": "pcs/h for converting equipment; m/min for corrugators."
        }
      },
      "additionalProperties": false
    },

    "LogisticsStatusItem": {
      "description": "Status of one logistics conveyor or buffer section. Quantities describe current contents at observation time, not cumulative throughput. stacks counts physical stacks.",
      "type": "object",
      "required": ["statusId"],
      "properties": {
        "name":          { "type": ["string", "null"] },
        "streamId":      { "type": ["string", "null"] },
        "statusId":      { "type": "string" },
        "movement": {
          "type": "string",
          "enum": ["running", "stopped", "moving-left", "moving-right", "moving-forward", "moving-backward", "rotate-clockwise", "rotate-counterclockwise"],
          "description": "Optional movement status, subject to supplier/customer agreement. When provided, use the defined values; see message-properties.md."
        },
        "articleRef":    { "type": ["string", "null"] },
        "area":          { "$ref": "#/$defs/Quantity" },
        "weight":        { "$ref": "#/$defs/Quantity" },
        "runningLength": { "$ref": "#/$defs/Quantity" },
        "pieces":        { "$ref": "#/$defs/Quantity" },
        "stacks":        { "$ref": "#/$defs/Quantity" }
      },
      "additionalProperties": false
    },

    "DiePressureItem": {
      "description": "Die pressure at a given time during production of the job, in kN.",
      "type": "object",
      "allOf": [{ "$ref": "#/$defs/Quantity" }],
      "properties": { "uom": { "const": "kN" } }
    }

  },

  "properties": {

    "speed": {
      "oneOf": [
        { "$ref": "#/$defs/Speed" },
        { "type": "null" }
      ],
      "description": "Current and target speed. Null when unavailable or not applicable. Zero is an actual reported speed, not a missing value."
    },

    "energy": {
      "type": "array",
      "items": { "$ref": "#/$defs/EnergyItem" },
      "description": "Energy and resource consumption counters. One entry per stream or job."
    },

    "inputs": {
      "type": "array",
      "items": { "$ref": "#/$defs/StreamItem" },
```

```
      "description": "Consumed material streams (raw material infeed). Scope: roll stand to stacker for corrugators; pre-prefeeder to post-palletizer for converting."
    },

    "outputs": {
      "type": "array",
      "items": { "$ref": "#/$defs/StreamItem" },
      "description": "Produced material streams (finished goods outfeed)."
    },

    "logisticsStatus": {
      "type": "array",
      "items": { "$ref": "#/$defs/LogisticsStatusItem" },
      "description": "Optional: status of logistics conveyor and buffer sections."
    },

    "diePressure": {
      "$ref": "#/$defs/DiePressureItem",
      "description": "Optional: die pressure reading for flat-bed die cutters having such option, in kN."
    }

  },

  "additionalProperties": false
}
```

1.7 quality.schema.json

Source: schemas/quality.schema.json

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "cosi:quality-payload:1.0",
  "title": "COSI -- Quality Payload",
  "description": "Payload for quality messages (assetClass=quality). Contains per-job quality checks organised by production section. Sections not applicable to a machine type are omitted.",
  "type": "object",
  "required": ["jobRef", "articleRef", "taskRef", "setupRef", "qualityChecks"],

  "$defs": {

    "QualityCheck": {
      "description": "Every quality check uses a single uniform shape regardless of whether the value is numeric, boolean, or qualitative.",
      "type": "object",
      "required": ["errorType", "lowerLimit", "upperLimit", "uom", "checks"],
      "properties": {
        "errorType":  { "type": "string", "description": "Human-readable check label in English." },
        "lowerLimit": { "type": ["number", "null"], "description": "Lower tolerance bound. Null for non-numeric checks." },
        "upperLimit": { "type": ["number", "null"], "description": "Upper tolerance bound. Null for non-numeric checks." },
        "uom":        { "type": ["string", "null"], "description": "Unit of measure. Null when not applicable." },
        "checks": {
          "type": "array",
          "description": "Atomic quality checks with counted outcomes. Each entry describes a check; receiving applications build reports. This array is not a cumulative job report, and an empty array does not establish OK output.",
          "items": {
            "type": "object",
            "required": ["quantity", "uom", "result"],
            "properties": {
              "quantity":  { "type": "number", "description": "Number of pieces covered by this atomic check." },
              "uom":       { "type": "string", "description": "Unit of the count; use `pcs`" },
              "result":    { "type": "string", "enum": ["OK", "NOK"], "description": "Outcome for the counted pieces."},
              "ejected":   { "type": "boolean", "description": "Whether pieces covered by a NOK check were ejected; retain on NOK." },
              "exception": { "type": "string", "enum": ["lower-violation", "upper-violation"], "description": "Numeric bound violated on a NOK check. Omit when no numeric bound was violated, including qualitative NOK with null limits." }
            },
            "additionalProperties": false
          }
        }
      },
      "additionalProperties": false
    }

  },

  "properties": {

    "jobRef": {
      "type": ["string", "null"],
      "description": "Manufacturing order reference from ERP / MES."
    },

    "articleRef": {
      "type": ["string", "null"],
      "description": "Article / product specification reference."
    },
```

```
    "taskRef": {
      "type": ["string", "null"],
      "description": "Manufacturing step reference within the order."
    },

    "setupRef": {
      "type": ["string", "null"],
      "description": "Reference to setupSettings.setupId of the corrugator in the same site/machine context. Null when not applicable; distinct executions on one machine have distinct setup identifiers."
    },

    "qualityChecks": {
      "type": "object",
      "description": "Quality check results grouped by production section. Omit sections that are not applicable to the reporting machine.",
      "properties": {

        "corrugating": {
          "type": "object",
          "description": "Checks applicable to corrugating / corrugator output.",
          "properties": {
            "warp":             { "$ref": "#/$defs/QualityCheck" },
            "humidity":         { "$ref": "#/$defs/QualityCheck" },
            "temperature":      { "$ref": "#/$defs/QualityCheck" },
            "caliper":          { "$ref": "#/$defs/QualityCheck" },
            "paperQuality":     { "$ref": "#/$defs/QualityCheck" },
            "reelQuality":      { "$ref": "#/$defs/QualityCheck" },
            "paperAlignment":   { "$ref": "#/$defs/QualityCheck" },
            "printRegister":    { "$ref": "#/$defs/QualityCheck" },
            "blistersAndGluing":{ "$ref": "#/$defs/QualityCheck" },
            "cutQuality":       { "$ref": "#/$defs/QualityCheck" },
            "scoring":          { "$ref": "#/$defs/QualityCheck" },
            "format":           { "$ref": "#/$defs/QualityCheck" }
          },
          "additionalProperties": false
        },

        "foldingGluing": {
          "type": "object",
          "description": "Checks applicable to folder-gluer output.",
          "properties": {
            "gapSize":    { "$ref": "#/$defs/QualityCheck" },
            "glueApplied":{ "$ref": "#/$defs/QualityCheck" },
            "scoring":    { "$ref": "#/$defs/QualityCheck" },
            "fishtailing":{ "$ref": "#/$defs/QualityCheck" },
            "wasteFree":  { "$ref": "#/$defs/QualityCheck" }
          },
          "additionalProperties": false
        },

        "dieCutting": {
          "type": "object",
          "description": "Checks applicable to die-cutting output.",
          "properties": {
            "missingLinesCuttingCreasing": { "$ref": "#/$defs/QualityCheck" },
            "creasingCracking":            { "$ref": "#/$defs/QualityCheck" },
            "dimensions":                  { "$ref": "#/$defs/QualityCheck" }
          },
          "additionalProperties": false
        },

        "logistics": {
          "type": "object",
          "description": "Checks applicable to pallet / stack output.",
          "properties": {
            "stackAccuracy": { "$ref": "#/$defs/QualityCheck" },
            "stackHeight":   { "$ref": "#/$defs/QualityCheck" },
            "elephantFoot":  { "$ref": "#/$defs/QualityCheck" },
            "labelsPosition":{ "$ref": "#/$defs/QualityCheck" }
          },
          "additionalProperties": false
        },
```

```
        "printing": {
          "type": "object",
          "description": "Checks per print colour station.",
          "required": ["colors"],
          "properties": {
            "colors": {
              "type": "array",
              "minItems": 1,
              "items": {
                "type": "object",
                "required": ["colorId"],
                "properties": {
                  "colorId":          { "type": "string", "description": "Identifies the
                  ↪  print station (e.g., 'station-1', 'C', 'M')." },
                  "printingDefects":  { "$ref": "#/$defs/QualityCheck" },
                  "registerAccuracy": { "$ref": "#/$defs/QualityCheck" },
                  "colorAccuracy":    { "$ref": "#/$defs/QualityCheck" },
                  "inkTemperature":   { "$ref": "#/$defs/QualityCheck" },
                  "inkViscosity":     { "$ref": "#/$defs/QualityCheck" },
                  "inkPH":            { "$ref": "#/$defs/QualityCheck" }
                },
                "additionalProperties": false
              }
            }
          },
          "additionalProperties": false
        }

      },
      "additionalProperties": false
    }

  },

  "additionalProperties": false
}
```

1.8 setup-converting.schema.json

Source: schemas/setup-converting.schema.json

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "cosi:setup-converting-payload:1.0",
  "title": "COSI -- Setup Converting Payload",
  "description": "Payload for setup-converting messages (assetClass=setup-converting).
  ↪  Carries job, article, material, machine settings and periphery device setup for one
  ↪  converting machine job. ERP/MES sends setup instructions; the machine publishes changed
  ↪  setup settings back to ERP/MES, including operator edits (S2).",
  "type": "object",
  "required": ["jobSettings", "articleSettings", "rawMaterial", "machineSettings",
  ↪  "periphery"],

  "$defs": {

    "Quantity": { "$ref": "cosi:common-defs:1.0#/$defs/Quantity" },

    "QuantityMM": { "$ref": "cosi:common-defs:1.0#/$defs/QuantityMM" },

    "Score": {
      "description": "One scoring or creasing knife definition.",
      "type": "object",
      "required": ["scoreId", "scoreType", "position"],
      "properties": {
        "scoreId":   { "type": "integer", "minimum": 1 },
        "scoreType": { "type": ["string", "null"], "description": "Open enum -- e.g.,
        ↪  'V-cut', 'perf', 'half-cut'. Null for inactive scores." },
        "position":  { "$ref": "#/$defs/QuantityMM", "description": "Knife position from
        ↪  sheet edge in mm." }
      },
      "additionalProperties": false
    },

    "FlexoPrintSettings": {
      "type": "object",
      "required": ["active", "color", "inkName", "plateId", "position", "register",
      ↪  "cylinderCentering", "cylinderCalibration", "aniloxCalibration", "useDryer",
      ↪  "plateHeight"],
      "properties": {
        "active":              { "type": "boolean" },
        "color": {
          "type": "object",
          "required": ["value", "uom"],
          "properties": {
            "value": { "type": "string", "description": "Colour reference as a string --
            ↪  e.g., ARGB hex, Pantone code, or an ERP colour/SKU reference." },
            "uom":   { "type": "string", "description": "Open enum -- colour reference scheme
            ↪  (e.g., 'ARGB', 'Pantone', 'SKU')." }
          },
          "additionalProperties": false,
          "description": "Ink colour reference; the uom names the reference scheme."
        },
        "inkName":             { "type": "string" },
        "plateId":             { "type": ["string", "null"], "description": "Printing plate
        ↪  reference. Null when no plate is mounted." },
        "position":            { "type": "object", "allOf": [{ "$ref": "#/$defs/Quantity" }],
        ↪  "properties": { "uom": { "const": "station" } }, "description": "Print station
        ↪  number with uom='station'." },
        "register":            { "$ref": "#/$defs/QuantityMM", "description": "Longitudinal
        ↪  register offset in mm." },
        "cylinderCentering":   { "$ref": "#/$defs/QuantityMM" },
        "cylinderCalibration": { "$ref": "#/$defs/QuantityMM" },
        "aniloxCalibration":   { "$ref": "#/$defs/QuantityMM" },
        "useDryer":            { "type": "boolean" },
        "plateHeight":         { "$ref": "#/$defs/QuantityMM", "description": "Plate/sleeve
        ↪  height in mm." }
      },
      "additionalProperties": false
    },
```

```
    "DigitalInkjetSettings": {
      "type": "object",
      "required": ["active", "printFileUrl", "printWidth", "printHeight", "numberOfCopies", "colorCorrection", "resolution", "substrate", "orientation", "leadIn", "leadOut"],
      "properties": {
        "active":          { "type": "boolean" },
        "printFileUrl":    { "type": "string", "description": "URI to the print file." },
        "printWidth":      { "$ref": "#/$defs/QuantityMM" },
        "printHeight":     { "$ref": "#/$defs/QuantityMM" },
        "numberOfCopies":  { "type": "integer", "minimum": 0 },
        "colorCorrection": { "type": ["string", "null"] },
        "resolution":      { "type": "object", "allOf": [{ "$ref": "#/$defs/Quantity" }], "properties": { "uom": { "const": "dpi" } }, "description": "Print resolution with uom='dpi'." },
        "substrate":       { "type": ["string", "null"] },
        "orientation":     { "type": "string", "description": "Open enum -- e.g., 'TOP', 'BOTTOM'." },
        "leadIn":          { "$ref": "#/$defs/QuantityMM" },
        "leadOut":         { "$ref": "#/$defs/QuantityMM" }
      },
      "additionalProperties": false
    },

    "InfeedSettings": {
      "type": "object",
      "required": ["active", "feedType", "inputWidth", "inputLength", "numberIn"],
      "properties": {
        "active":   { "type": "boolean" },
        "feedType": { "type": ["string", "null"], "description": "Open enum -- e.g., 'vacuum', 'gripper'. Null when no special feed system is used." },
        "inputWidth":   { "$ref": "#/$defs/QuantityMM", "description": "Input sheet width in mm." },
        "inputLength":  { "$ref": "#/$defs/QuantityMM", "description": "Input sheet length in mm." },
        "numberIn": { "type": "number", "minimum": 0 }
      },
      "additionalProperties": false
    },

    "DieCutterSettings": {
      "type": "object",
      "required": ["active", "cutterType", "upsWidth", "upsLength", "toolRef", "diePressure", "cuttingForce", "offsetScoring", "scores"],
      "properties": {
        "stationNumber": { "type": "integer"},
        "active":        { "type": "boolean" },
        "cutterType":    { "type": "string", "description": "Open enum -- e.g., 'rotary', 'flatbed'." },
        "upsWidth":      { "type": "integer", "minimum": 1, "description": "Number of ups across sheet width." },
        "upsLength":     { "type": "integer", "minimum": 1, "description": "Number of ups along sheet length." },
        "toolRef":       { "type": ["string", "null"], "description": "Die-cutting tool reference. Null when inactive." },
        "diePressure":   { "type": "object", "allOf": [{ "$ref": "#/$defs/Quantity" }], "properties": { "uom": { "const": "kN" } }, "description": "Die pressure in kN." },
        "cuttingForce":  { "type": "object", "allOf": [{ "$ref": "#/$defs/Quantity" }], "properties": { "uom": { "const": "kN" } }, "description": "Cutting force in kN." },
        "cuttingDepth":  { "$ref": "#/$defs/QuantityMM", "description": "Cutting depth in mm." },
        "offsetScoring": { "$ref": "#/$defs/QuantityMM", "description": "Scoring offset in mm." },
        "scores": {
          "type": "array",
          "items": { "$ref": "#/$defs/Score" }
        }
      },
      "additionalProperties": false
    },

    "FolderGluerSettings": {
```

```
      "type": "object",
      "required": ["active", "glueType", "glueStyle", "glueAmount"],
      "properties": {
        "active":     { "type": "boolean" },
        "glueType":   { "type": "integer", "description": "Glue type code (machine-specific integer)." },
        "glueStyle":  { "type": "string",  "description": "Open enum -- e.g., 'GI' (glue inside), 'GO' (glue outside), '3P' (three-point)." },
        "glueAmount": { "type": "object", "allOf": [{ "$ref": "#/$defs/Quantity" }], "properties": { "uom": { "const": "g/m²" } }, "description": "Glue application amount in g/m²." }
      },
      "additionalProperties": false
    },

    "StitcherSettings": {
      "type": "object",
      "required": ["active", "stitchType"],
      "properties": {
        "active":     { "type": "boolean" },
        "stitchType": { "type": "integer", "description": "Stitch type code (machine-specific integer)." }
      },
      "additionalProperties": false
    },

    "TaperSettings": {
      "type": "object",
      "required": ["active", "tapeType"],
      "properties": {
        "active":   { "type": "boolean" },
        "tapeType": { "type": "string", "description": "Open enum -- tape product type." }
      },
      "additionalProperties": false
    },

    "BundleStrapperSettings": {
      "type": "object",
      "required": ["active", "itemsPerBundle", "bundleWidth", "bundleHeight", "bundlePositionLateral"],
      "properties": {
        "active":               { "type": "boolean" },
        "itemsPerBundle":       { "type": "object", "allOf": [{ "$ref": "#/$defs/Quantity" }], "properties": { "uom": { "const": "pcs" } }, "description": "Number of items per bundle (uom='pcs')." },
        "bundleWidth":          { "$ref": "#/$defs/QuantityMM" },
        "bundleHeight":         { "$ref": "#/$defs/QuantityMM" },
        "bundlePositionLateral": { "$ref": "#/$defs/QuantityMM", "description": "Lateral centre position of the bundle in mm." }
      },
      "additionalProperties": false
    },

    "PalletizerSettings": {
      "type": "object",
      "required": ["active", "palletType", "palletLength", "palletWidth", "maxHeight", "itemsPerPallet", "numberOfLayers", "packetsPerLayer", "compositionRef", "interleaves", "strapping", "wrapping", "counterPallet", "coverSheet", "bottomSheet", "cornerProtection", "label"],
      "properties": {
        "active":          { "type": "boolean" },
        "palletType":      { "type": "integer", "description": "Pallet type code (machine-specific integer)." },
        "palletLength":    { "$ref": "#/$defs/QuantityMM" },
        "palletWidth":     { "$ref": "#/$defs/QuantityMM" },
        "maxHeight":       { "$ref": "#/$defs/QuantityMM", "description": "Maximum allowed pallet height in mm." },
        "itemsPerPallet":  { "type": "object", "allOf": [{ "$ref": "#/$defs/Quantity" }], "properties": { "uom": { "const": "pcs" } }, "description": "Total items per pallet (uom='pcs')." },
        "numberOfLayers":  { "type": "integer", "minimum": 0 },
        "packetsPerLayer": { "type": "integer", "minimum": 0 },
        "compositionRef":  { "type": ["string", "null"], "description": "Reference to the pallet composition pattern. Null if not defined." },
```

```
        "interleaves":     { "type": "boolean", "description": "True when interleave sheets are placed between pallet layers." },
        "strapping": {
          "type": "object",
          "required": ["active", "positions", "numberOfStraps", "pressure", "programId"],
          "properties": {
            "active":         { "type": "boolean" },
            "positions":      { "type": "array", "items": { "type": "number" }, "description": "Strap positions in mm from pallet edge." },
            "numberOfStraps": { "type": "integer", "minimum": 0 },
            "pressure":       { "type": "object", "allOf": [{ "$ref": "#/$defs/Quantity" }], "properties": { "uom": { "const": "kN" } }, "description": "Strap tension in kN." },
            "programId":      { "type": ["string", "null"], "description": "Strapping program reference. Null when not defined." }
          },
          "additionalProperties": false
        },
        "wrapping": {
          "type": "object",
          "required": ["active", "programId", "wrapType"],
          "properties": {
            "active":    { "type": "boolean" },
            "programId": { "type": ["string", "null"], "description": "Wrapping program identifier. Null when not defined." },
            "wrapType":  { "type": "integer", "description": "Wrap type code (machine-specific integer)." }
          },
          "additionalProperties": false
        },
        "counterPallet":    { "type": "boolean", "description": "True when a counter-pallet is placed on top." },
        "coverSheet":       { "type": "boolean" },
        "bottomSheet":      { "type": "boolean" },
        "cornerProtection": { "type": "boolean" },
        "label": {
          "type": "object",
          "required": ["position", "data"],
          "properties": {
            "position": { "type": ["string", "null"], "description": "Label placement position (open enum -- e.g., 'OS-front'). Null if not defined." },
            "data":     { "type": ["string", "null"], "description": "Label content or template reference. Null if not defined." }
          },
          "additionalProperties": false
        }
      },
      "additionalProperties": false
    }

  },

  "properties": {

    "customProps": {
      "$ref": "cosi:common-defs:1.0#/$defs/CustomProps"
    },

    "quickSetId": {
      "type": ["string", "null"],
      "description": "Reference to a saved quick-set template. Null when not used."
    },

    "jobSettings": {
      "type": "object",
      "required": ["jobRef", "orderRef", "customer", "dueDate", "quantity", "autoStopFeed"],
      "properties": {
        "jobRef":   { "type": "string", "description": "Manufacturing order reference from ERP / MES." },
        "orderRef": { "type": ["string", "null"], "description": "Sub-order reference within the job. Null when not applicable." },
        "customer": { "type": ["string", "null"] },
        "dueDate":  { "type": ["string", "null"], "format": "date-time", "description": "Production due date -- ISO 8601 UTC. Null when not planned." },
```

```
        "quantity": {
          "type": "object",
          "required": ["uom"],
          "properties": {
            "uom":       { "type": "string", "const": "pcs" },
            "scheduled": { "type": ["number", "null"], "minimum": 0 }
          },
          "additionalProperties": false,
          "description": "Qualified quantity group: scheduled = planned quantity (production actuals are not part of setup messages, see ADR-000005)."
        },
        "autoStopFeed": { "type": "boolean", "description": "True when the feeder auto-stops at the scheduled quantity." }
      },
      "additionalProperties": false
    },

    "articleSettings": {
      "type": "object",
      "required": ["articleRef", "boxType", "boxWidth", "boxLength", "boxHeight", "glueTab", "register", "panels"],
      "properties": {
        "articleRef": { "type": "string", "description": "Article / product specification reference." },
        "boxType":    { "type": "object", "allOf": [{ "$ref": "#/$defs/Quantity" }], "properties": { "uom": { "const": "FEFCO" } }, "description": "FEFCO style code as integer value with uom='FEFCO'." },
        "boxWidth":   { "$ref": "#/$defs/QuantityMM", "description": "Box internal width in mm." },
        "boxLength":  { "$ref": "#/$defs/QuantityMM", "description": "Box internal length in mm." },
        "boxHeight":  { "$ref": "#/$defs/QuantityMM", "description": "Box internal height in mm." },
        "glueTab":    { "$ref": "#/$defs/QuantityMM", "description": "Glue tab width in mm." },
        "register":   { "$ref": "#/$defs/QuantityMM", "description": "Print register offset in mm." },
        "panels": {
          "type": "array",
          "minItems": 1,
          "items": {
            "type": "object",
            "required": ["panelId", "value", "uom"],
            "properties": {
              "panelId": { "type": "string", "description": "Panel label (e.g., 'A', 'B', 'C', 'D', 'A1')." },
              "value":   { "type": "number" },
              "uom":     { "type": "string", "const": "mm" }
            },
            "additionalProperties": false
          },
          "description": "Panel width breakdown. Panel sequence follows FEFCO notation."
        }
      },
      "additionalProperties": false
    },

    "rawMaterial": {
      "type": "object",
      "required": ["flute", "grade", "boardThickness", "surfaceType", "crossFlute", "papers"],
      "properties": {
        "flute":          { "type": "string", "description": "Open enum -- flute type (e.g., 'B', 'C', 'BC', 'E')." },
        "grade":          { "type": "string", "description": "Board grade description." },
        "boardThickness": { "$ref": "#/$defs/QuantityMM", "description": "Nominal board caliper in mm." },
        "surfaceType":    { "type": ["string", "null"], "description": "Open enum -- surface treatment (e.g., 'white-top', 'coated'). Null when not applicable." },
        "crossFlute":     { "type": "boolean", "description": "True when the board is run cross-flute orientation." },
        "papers": {
          "type": "array",
          "minItems": 1,
```

```
          "items": {
            "type": "object",
            "required": ["layer", "gsm", "paperType"],
            "properties": {
              "layer":     { "type": "string", "description": "Open enum -- layer position (e.g., 'liner-top', 'fluting', 'liner-bottom')." },
              "gsm":       { "type": "object", "allOf": [{ "$ref": "#/$defs/Quantity" }], "properties": { "uom": { "const": "g/m²" } }, "description": "Paper grammage in g/m²." },
              "paperType": { "type": ["string", "null"], "description": "Open enum -- paper type description. Null when not specified." }
            },
            "additionalProperties": false
          }
        }
      },
      "additionalProperties": false
    },

    "machineSettings": {
      "type": "object",
      "required": ["taskRef", "stageNo", "outputWidth", "outputLength", "speed", "setupDuration", "infeedSettings"],
      "properties": {
        "taskRef":      { "type": ["string", "null"], "description": "Manufacturing step reference within the order. Null when not assigned." },
        "stageNo":      { "type": "integer", "minimum": 0, "description": "Production stage number within a multi-stage job." },
        "outputWidth":  { "$ref": "#/$defs/QuantityMM", "description": "Output blank width in mm." },
        "outputLength": { "$ref": "#/$defs/QuantityMM", "description": "Output blank length in mm." },
        "infeedSettings": { "$ref": "#/$defs/InfeedSettings" },
        "speed": {
          "type": "object",
          "required": ["uom"],
          "properties": {
            "uom":       { "type": "string", "const": "pcs/h" },
            "target":    { "type": ["number", "null"], "minimum": 0 },
            "scheduled": { "type": ["number", "null"], "minimum": 0 }
          },
          "additionalProperties": false,
          "description": "Qualified speed group in pcs/h."
        },
        "setupDuration": {
          "type": "object",
          "required": ["uom"],
          "properties": {
            "uom":       { "type": "string", "const": "s" },
            "target":    { "type": ["number", "null"], "minimum": 0 },
            "scheduled": { "type": ["number", "null"], "minimum": 0 }
          },
          "additionalProperties": false,
          "description": "Qualified setup duration group in seconds."
        }
      },
      "additionalProperties": false
    },

    "periphery": {
      "type": "array",
      "items": {
        "type": "object",
        "required": ["deviceId", "deviceType", "settings"],
        "properties": {
          "deviceId":   { "type": "string", "description": "Machine-internal device identifier." },
          "deviceType": { "type": "string", "enum": ["flexo-print", "digital-inkjet", "die-cutter", "folder-gluer", "stitcher", "taper", "bundle-strapper", "palletizer"] },
          "settings":   { "type": "object" }
        },
        "oneOf": [
          {
```

```
            "properties": {
              "deviceType": { "const": "flexo-print" },
              "settings":   { "$ref": "#/$defs/FlexoPrintSettings" }
            }
          },
          {
            "properties": {
              "deviceType": { "const": "digital-inkjet" },
              "settings":   { "$ref": "#/$defs/DigitalInkjetSettings" }
            }
          },
          {
            "properties": {
              "deviceType": { "const": "die-cutter" },
              "settings":   { "$ref": "#/$defs/DieCutterSettings" }
            }
          },
          {
            "properties": {
              "deviceType": { "const": "folder-gluer" },
              "settings":   { "$ref": "#/$defs/FolderGluerSettings" }
            }
          },
          {
            "properties": {
              "deviceType": { "const": "stitcher" },
              "settings":   { "$ref": "#/$defs/StitcherSettings" }
            }
          },
          {
            "properties": {
              "deviceType": { "const": "taper" },
              "settings":   { "$ref": "#/$defs/TaperSettings" }
            }
          },
          {
            "properties": {
              "deviceType": { "const": "bundle-strapper" },
              "settings":   { "$ref": "#/$defs/BundleStrapperSettings" }
            }
          },
          {
            "properties": {
              "deviceType": { "const": "palletizer" },
              "settings":   { "$ref": "#/$defs/PalletizerSettings" }
            }
          }
        ],
        "additionalProperties": false
      },
      "description": "Periphery devices on this converting line. Each device appears once; inactive devices include active=false in settings."
    }

  },

  "additionalProperties": false
}
```

1.9 setup-corrugator.schema.json

Source: schemas/setup-corrugator.schema.json

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "cosi:setup-corrugator-payload:1.0",
  "title": "COSI -- Setup Corrugator Payload",
  "description": "Payload for setup-corrugator messages (assetClass=setup-corrugator).
  ↪  Carries one corrugator setup with board grade, paper specs, per-order cutting/scoring
  ↪  layout, and periphery device settings. ERP/MES sends setup instructions; the machine
  ↪  publishes changed setup settings back to ERP/MES, including operator edits (S2).",
  "type": "object",
  "required": ["setupSettings", "boardGradeSettings", "orders", "periphery"],

  "$defs": {

    "Quantity": { "$ref": "cosi:common-defs:1.0#/$defs/Quantity" },

    "QuantityMM": { "$ref": "cosi:common-defs:1.0#/$defs/QuantityMM" },

    "PaperSpec": {
      "description": "Paper specification for one roll position.",
      "type": "object",
      "required": ["paperId", "gsm", "caliper", "width", "requiredFSC"],
      "properties": {
        "paperId":     { "type": ["string", "null"], "description": "Paper grade or article
        ↪  identifier. Null when the position is inactive." },
        "gsm":         { "type": "object", "allOf": [{ "$ref": "#/$defs/Quantity" }],
        ↪  "properties": { "uom": { "const": "g/m²" } }, "description": "Paper grammage in
        ↪  g/m²." },
        "caliper":     { "$ref": "#/$defs/QuantityMM", "description": "Paper caliper in mm."
        ↪  },
        "width":       { "$ref": "#/$defs/QuantityMM", "description": "Reel width in mm." },
        "requiredFSC": {
          "type": ["string", "null"],
          "description": "Required FSC certification level (open enum -- e.g., '100%', '75%',
          ↪  'Recycled'). Null when FSC is not required."
        }
      },
      "additionalProperties": false
    },

    "ProcessAdds": {
      "description": "Process additions applied to a paper web.",
      "type": "object",
      "required": ["waterResistant", "whiteTop", "wax", "coating"],
      "properties": {
        "waterResistant": { "type": "boolean" },
        "whiteTop":       { "type": "boolean" },
        "wax":            { "type": "boolean" },
        "coating":        { "type": "boolean" }
      },
      "additionalProperties": false
    },

    "Preprint": {
      "description": "Preprint register mark settings for one corrugator order.",
      "type": "object",
      "required": ["active", "markDefinition", "mark1Width", "gap1Width", "mark2Width",
      ↪  "gap2Width", "mark3Width", "windowWidth", "markToEdge", "markOffset", "printSide",
      ↪  "colorContrast"],
      "properties": {
        "active":         { "type": "boolean" },
        "markDefinition": { "type": "integer", "minimum": 0 },
        "mark1Width":     { "$ref": "#/$defs/QuantityMM" },
        "gap1Width":      { "$ref": "#/$defs/QuantityMM" },
        "mark2Width":     { "$ref": "#/$defs/QuantityMM" },
        "gap2Width":      { "$ref": "#/$defs/QuantityMM" },
        "mark3Width":     { "$ref": "#/$defs/QuantityMM" },
        "windowWidth":    { "$ref": "#/$defs/QuantityMM" },
        "markToEdge":     { "$ref": "#/$defs/QuantityMM" },
```

```
        "markOffset":     { "$ref": "#/$defs/QuantityMM" },
        "printSide":      { "type": "string", "description": "Open enum -- e.g., 'bottom', 'top'." },
        "colorContrast":  { "type": "number" }
      },
      "additionalProperties": false
    },

    "ScoringProfile": {
      "description": "Scoring tool profile definition.",
      "type": "object",
      "required": ["profileId", "profileType", "offset", "depth"],
      "properties": {
        "profileId":   { "type": "integer", "minimum": 1 },
        "profileType": { "type": ["string", "null"], "description": "Tool profile type (e.g., 'V-cut'). Null for inactive profiles." },
        "offset":      { "type": "integer", "description": "Direction code: -1 = left, 0 = centre, 1 = right, 2 = special." },
        "depth":       { "$ref": "#/$defs/QuantityMM", "description": "Scoring depth in mm." }
      },
      "additionalProperties": false
    },

    "ScoringPosition": {
      "description": "One scoring knife position.",
      "type": "object",
      "required": ["position", "value", "uom", "profileId"],
      "properties": {
        "position":  { "type": "integer", "minimum": 1, "maximum": 24 },
        "value":     { "type": "number", "minimum": 0, "description": "Relative position for 1 order for 1 out." },
        "uom":       { "type": "string", "const": "mm" },
        "profileId": { "type": "integer", "minimum": 1, "description": "References a profileId in scoringProfiles." }
      },
      "additionalProperties": false
    },

    "SingleFacerSettings": {
      "type": "object",
      "required": ["active", "linerPaper", "mediumPaper", "fluteType", "processAdds"],
      "properties": {
        "active":       { "type": "boolean" },
        "linerPaper":   { "$ref": "#/$defs/PaperSpec" },
        "mediumPaper":  { "$ref": "#/$defs/PaperSpec" },
        "fluteType":    { "type": ["string", "null"], "description": "Flute profile (e.g., 'B', 'C'). Null when the single facer is inactive." },
        "processAdds":  { "$ref": "#/$defs/ProcessAdds" }
      },
      "additionalProperties": false
    },

    "DoubleBackerSettings": {
      "type": "object",
      "required": ["active", "linerPaper", "processAdds"],
      "properties": {
        "active":      { "type": "boolean" },
        "linerPaper":  { "$ref": "#/$defs/PaperSpec" },
        "processAdds": { "$ref": "#/$defs/ProcessAdds" }
      },
      "additionalProperties": false
    },

    "FanfoldSettings": {
      "type": "object",
      "required": ["active", "orderIndex", "numberOfFolds", "creaseLength"],
      "properties": {
        "active":        { "type": "boolean" },
        "orderIndex":    { "type": "integer", "minimum": 0, "description": "Index of the order this fanfold device serves (0 = unassigned)." },
        "numberOfFolds": { "type": "integer", "minimum": 0 },
        "creaseLength":  { "$ref": "#/$defs/QuantityMM" }
      },
```

```
      "additionalProperties": false
    },

    "TapeSettings": {
      "type": "object",
      "required": ["active", "orderIndex", "tapeType", "positions"],
      "properties": {
        "active":     { "type": "boolean" },
        "orderIndex": { "type": "integer", "minimum": 0 },
        "tapeType":   { "type": "string", "description": "Open enum -- tape material or product type." },
        "positions":  {
          "type": "array",
          "items": { "$ref": "#/$defs/QuantityMM" },
          "description": "Lateral tape application positions in mm from web edge."
        }
      },
      "additionalProperties": false
    }

  },

  "properties": {

    "customProps": {
      "$ref": "cosi:common-defs:1.0#/$defs/CustomProps"
    },

    "setupSettings": {
      "type": "object",
      "required": ["setupId", "webWidth", "leftTrim", "rightTrim", "speed", "control1Speed", "control2Speed", "runningLength"],
      "properties": {
        "setupId": {
          "type": "string",
          "minLength": 1,
          "description": "Identifier of one planned execution, unique within site/machine. Revisions and continuation after interruption preserve it; a separate execution gets a distinct identifier. quality.setupRef references it in the same context."
        },
        "webWidth":   { "$ref": "#/$defs/QuantityMM", "description": "Full corrugator web width." },
        "leftTrim":   { "$ref": "#/$defs/QuantityMM", "description": "Left edge trim in mm." },
        "rightTrim":  { "$ref": "#/$defs/QuantityMM", "description": "Right edge trim in mm." },
        "speed": {
          "type": "object",
          "required": ["uom"],
          "properties": {
            "uom":     { "type": "string", "const": "m/min" },
            "target":  { "type": ["number", "null"], "minimum": 0 }
          },
          "additionalProperties": false,
          "description": "Qualified speed group. target = reference speed supplied by ERP/MES for this setup according to the plant's business rules. COSI does not prescribe how the baseline is calculated."
        },
        "control1Speed": { "$ref": "#/$defs/Quantity", "description": "Speed setting for control point 1 in m/min." },
        "control2Speed": { "$ref": "#/$defs/Quantity", "description": "Speed setting for control point 2 in m/min." },
        "runningLength": {
          "type": "object",
          "required": ["uom"],
          "properties": {
            "uom":       { "type": "string", "const": "m" },
            "scheduled": { "type": ["number", "null"], "minimum": 0 }
          },
          "additionalProperties": false,
          "description": "Requested running length for this execution in metres; not actual production."
        }
```

```
      },
      "additionalProperties": false
    },

    "boardGradeSettings": {
      "type": "object",
      "required": ["boardGradeRef", "fluteType", "caliper"],
      "properties": {
        "boardGradeRef": { "type": "string", "description": "Reference to the board grade specification in ERP / MES." },
        "fluteType":     { "type": "string", "description": "Open enum -- flute combination (e.g., 'B', 'C', 'BC', 'E')." },
        "caliper":       { "$ref": "#/$defs/QuantityMM", "description": "Target board caliper in mm." }
      },
      "additionalProperties": false
    },


    "orders": {
      "type": "array",
      "minItems": 1,
      "items": {
        "type": "object",
        "required": ["orderType", "l2rSequenceNumber"],
        "properties": {
          "customProps": { "$ref": "cosi:common-defs:1.0#/$defs/CustomProps" },
          "orderType": {
            "type": "string",
            "enum": ["production", "waste-run", "side-order"],
            "description": "'production' = standard corrugator order. 'waste-run' = trim strip too wide for the trim system. 'side-order' = productive order cut from a side strip."
          },
          "orderRef":          { "type": "string", "description": "Manufacturing order reference from ERP / MES." },
          "partRef":           { "type": "string", "description": "Part reference within the order." },
          "articleRef":        { "type": "string", "description": "Article / product specification reference." },
          "customerName":      { "type": "string" },
          "processingMachine": { "type": "string", "description": "Identifier of the converting machine that will process this order." },
          "destinationLine":   { "type": "integer", "minimum": 0, "description": "Destination output line number." },
          "sheetWidth":        { "$ref": "#/$defs/QuantityMM", "description": "Cut sheet width." },
          "sheetLength":       { "$ref": "#/$defs/QuantityMM", "description": "Cut sheet length." },
          "cuts": {
            "type": "object",
            "required": ["uom"],
            "properties": {
              "uom":    { "type": "string", "const": "cuts" },
              "target": { "type": ["number", "null"], "minimum": 0 }
            },
            "additionalProperties": false,
            "description": "Requested number of cross-cuts for this order; setup instructions do not establish actual output."
          },
          "numberOfOuts":      { "type": "integer", "minimum": 1, "description": "Number of sheets cut side by side from the web width." },
          "cutoffAllocation":  { "type": "integer", "minimum": 0, "description": "Cut-off sheet allocation offset." },
          "l2rSequenceNumber": { "type": "integer", "minimum": 0, "description": "Left-to-right sequence number of the orders defining the layout on the machine. See glossary for definition." },
          "preprint":        { "$ref": "#/$defs/Preprint" },
          "scoringProfiles": {
            "type": "array",
            "minItems": 1,
            "items": { "$ref": "#/$defs/ScoringProfile" },
            "description": "Scoring tool profiles available for this order."
          },
```

```
          "scoringPositions": {
            "type": "array",
            "minItems": 24,
            "maxItems": 24,
            "items": { "$ref": "#/$defs/ScoringPosition" },
            "description": "Exactly 24 scoring knife positions. Unused positions carry value 0."
          },
          "stackHeight":       { "type": "object", "allOf": [{ "$ref": "#/$defs/Quantity" }], "properties": { "uom": { "const": "mm" } }, "description": "Requested physical stack height in millimetres." },
          "stackCount":        { "type": "object", "allOf": [{ "$ref": "#/$defs/Quantity" }], "properties": { "uom": { "const": "pcs" } }, "description": "Requested number of sheets per stack; not a count of physical stacks or actual output." },
          "lastRun":           { "type": "boolean", "description": "True when this is the final run for this order." },
          "dischargeDirection": { "type": "string", "description": "Open enum -- e.g., 'OS' (operator side), 'DS' (drive side)." },
          "intermediateLayer":  { "type": "boolean", "description": "True when an intermediate layer sheet is inserted between stacks." },
          "palletType":        { "type": "string", "description": "Open enum -- pallet type (e.g., 'Euro', 'Industrial')." }
        },
        "if": {
          "properties": {
            "orderType": { "const": "production" }
          }
        },
        "then": {
          "required": ["orderRef", "partRef", "articleRef", "customerName", "processingMachine", "destinationLine", "sheetWidth", "sheetLength", "cuts", "numberOfOuts", "cutoffAllocation", "preprint", "scoringProfiles", "scoringPositions", "stackHeight", "stackCount", "lastRun", "dischargeDirection", "intermediateLayer", "palletType"]
        },
        "else": {
          "required": ["sheetWidth", "sheetLength", "cutoffAllocation"]
        },
        "additionalProperties": false
      },
      "description": "Corrugator orders within this setup (typically 1-3), including production orders and optional side/waste strips in one shared array."
    },

    "periphery": {
      "type": "array",
      "items": {
        "type": "object",
        "required": ["deviceId", "deviceType", "settings"],
        "properties": {
          "deviceId":   { "type": "string", "description": "Machine-internal device identifier (e.g., 'SF1', 'DB')." },
          "deviceType": { "type": "string", "enum": ["single-facer", "double-backer", "fanfold", "tape"] },
          "settings":   { "type": "object" }
        },
        "oneOf": [
          {
            "properties": {
              "deviceType": { "const": "single-facer" },
              "settings":   { "$ref": "#/$defs/SingleFacerSettings" }
            }
          },
          {
            "properties": {
              "deviceType": { "const": "double-backer" },
              "settings":   { "$ref": "#/$defs/DoubleBackerSettings" }
            }
          },
          {
            "properties": {
              "deviceType": { "const": "fanfold" },
              "settings":   { "$ref": "#/$defs/FanfoldSettings" }
```

```
            }
          },
          {
            "properties": {
              "deviceType": { "const": "tape" },
              "settings":   { "$ref": "#/$defs/TapeSettings" }
            }
          }
        ],
        "additionalProperties": false
      },
      "description": "Periphery devices attached to the corrugator (single facers, double backer, fanfold, tape)."
    }

  },

  "additionalProperties": false
}
```

1.10 setup-response.schema.json

Source: schemas/setup-response.schema.json

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "cosi:setup-response-payload:1.0",
  "title": "COSI -- Setup Response Payload",
  "description": "Payload for setup-corrugator/setup-converting response messages.
  ↪  Correlation to the originating setup message is done via envelope correlationID, not
  ↪  via setup identifiers in payload.",
  "type": "object",
  "oneOf": [
    { "$ref": "#/$defs/ResponseOk" },
    { "$ref": "#/$defs/ResponseError" }
  ],

  "$defs": {

    "ErrorItem": {
      "description": "One error linked to a setup field. Manufacturer-specific error
      ↪  classification is intentionally open (no standardised code list).",
      "type": "object",
      "required": ["fieldRef", "description", "manufacturerCode"],
      "properties": {
        "fieldRef": {
          "type": "string",
          "minLength": 1,
          "pattern": "^(?:/(?:[^~/]|~[01])*)+$",
          "description": "RFC 6901 JSON Pointer rooted at the original full setup message
          ↪  identified by correlationID, e.g. /payload/orders/0/sheetWidth/value.
          ↪  Zero-based array indices; escape ~ as ~0 and / as ~1. JSON string form, not a
          ↪  URI fragment or dotted path."
        },
        "description": {
          "type": "string",
          "minLength": 1,
          "description": "Human-readable explanation of the error."
        },
        "manufacturerCode": {
          "type": "integer",
          "minimum": 0,
          "description": "Manufacturer-defined error code. Open value set by design."
        }
      },
      "additionalProperties": false
    },

    "ResponseOk": {
      "type": "object",
      "required": ["result"],
      "properties": {
        "result": {
          "type": "string",
          "const": "OK",
          "description": "Processing ok for the correlated setup message."
        }
      },
      "additionalProperties": false
    },

    "ResponseError": {
      "type": "object",
      "required": ["result", "errors"],
      "properties": {
        "result": {
          "type": "string",
          "const": "NOK",
          "description": "Processing failed for the correlated setup message; details in
          ↪  errors."
        },
        "errors": {
```

```
          "type": "array",
          "minItems": 1,
          "items": { "$ref": "#/$defs/ErrorItem" },
          "description": "One or multiple errors returned for this setup."
        }
      },
      "additionalProperties": false
    }

  }
}
```

2 Samples

2.1 lastwill/lastwill.json

Source: examples/lastwill/lastwill.json

```
{
  "messageID": "0e0632d3-34d2-55e3-a486-6e43c02b8c3d",
  "connectionId": "82fece39-6f80-5226-be75-9f2a6dc93907",
  "assetClass": "lastwill",
  "schemaId": "cosi:lastwill-payload:1.0",
  "messageType": "cos",
  "machineId": "corr-line-01",
  "timestamp": null,
  "status": "lastwill",
  "payload": {}
}
```

2.2 maintenance/maintenance-document.json

Source: examples/maintenance/maintenance-document.json


```
{
  "messageID":     "b9c8d7e6-f5a4-5b2c-9d0e-9f8a7b6c5d4e",
  "assetClass":    "maintenance-document",
  "schemaId":      "cosi:maintenance-document-payload:1.0",
  "messageType":   "onetime",
  "machineId":     "corr-line-01",
  "timestamp":     "2026-05-04T05:00:00.000Z",
  "status":        "maintenance",
  "payload": {

    "componentId": "SF1-PRES-ROLL-01",

    "manualDocs": [
      {
        "documentRef": {
          "uniqueRef":    "DOC-SF1-MAIN-001",
          "documentLink": "file:///manuals/corr-line-01/SF1/maintenance-pressure-roll.pdf"
        },
        "validFrom": "2025-06-01T00:00:00.000Z"
      },
      {
        "documentRef": {
          "uniqueRef":    "DOC-SF1-SAFETY-001",
          "documentLink": "file:///manuals/corr-line-01/SF1/safety-single-facer.pdf"
        },
        "validFrom": "2024-01-15T00:00:00.000Z"
      }
    ],

    "serviceTasks": [
      {
        "taskId":        "SRV-SF1-GREASE-BEARING",
        "frequency":     { "value": 500, "uom": "h" },
        "counterSource": "SF1 drive runtime hours",
        "documentRef": {
          "uniqueRef":    "DOC-SF1-GREASE-001",
          "documentLink": "file:///manuals/corr-line-01/SF1/grease-pressure-roll-bearing.pdf"
        }
      },
      {
        "taskId":        "SRV-SF1-REPLACE-BELT",
        "frequency":     { "value": 8000000, "uom": "cycles" },
        "counterSource": "SF1 corrugator cycle counter",
        "documentRef": {
          "uniqueRef":    "DOC-SF1-BELT-001",
          "documentLink": "file:///manuals/corr-line-01/SF1/replace-drive-belt.pdf"
        }
      },
      {
        "taskId":        "SRV-SF1-INSPECT-FLUTES",
        "frequency":     { "value": 2000, "uom": "h" },
        "counterSource": "SF1 drive runtime hours",
        "documentRef": {
          "uniqueRef":    "DOC-SF1-FLUTE-001",
          "documentLink": "file:///manuals/corr-line-01/SF1/inspect-fluting-roll.pdf"
        }
      },
      {
        "taskId":        "SRV-SF1-REPLACE-GLUE-ROLL",
        "frequency":     { "value": 1500000, "uom": "m²" },
        "counterSource": "SF1 board area counter",
        "documentRef": {
          "uniqueRef":    "DOC-SF1-GLUEROLL-001",
          "documentLink": "file:///manuals/corr-line-01/SF1/replace-glue-roll.pdf"
        }
      }
    ]

  }
```

```
}
```

2.3 maintenance/maintenance-measurement.json

Source: examples/maintenance/maintenance-measurement.json


```
{
  "messageID":     "c9d8e7f6-a5b4-5c2d-9e0f-9a8b7c6d5e4f",
  "assetClass":    "maintenance-measurement",
  "schemaId":      "cosi:maintenance-measurement-payload:1.0",
  "messageType":   "cos",
  "machineId":     "corr-line-01",
  "timestamp":     "2026-05-04T07:14:05.000Z",
  "status":        "fault",
  "payload": {

    "componentId":   "SF1-PRES-ROLL-01",
    "componentName": "SF1 Pressure Roll -- Drive-Side Bearing",

    "measuredValue": { "value": 8.7,  "uom": "mm/s" },
    "upperLimit":    { "value": 7.1,  "uom": "mm/s" },
    "lowerLimit":    { "value": null, "uom": null    },

    "failure": {
      "faultCode":     "VIB",
      "internalCode":  "SF1-VIB-DS-001",
      "descLocalLang": "Lagervibrationen überschreiten Grenzwert -- Lager prüfen",
      "descEN":        "Bearing vibration exceeds upper limit -- inspect bearing",
      "severity":      "warning"
    },

    "documentRef": {
      "uniqueRef":    "DOC-SF1-MAIN-001",
      "documentLink": "file:///manuals/corr-line-01/SF1/maintenance-pressure-roll.pdf"
    }

  }
}
```

2.4 performance/performance.json

Source: examples/performance/performance.json

```
{
  "messageID":     "3fa6f4e2-1c2d-5b3e-9a1f-2d7c8e5b0a4d",
  "connectionId": "82fece39-6f80-5226-be75-9f2a6dc93907",
  "assetClass":    "performance",
  "schemaId":      "cosi:performance-payload:1.0",
  "messageType":   "hbt",
  "machineId":     "corr-line-01",
  "timestamp":     "2026-05-04T06:30:00.000Z",
  "status":        "run",
  "payload": {
    "speed": {
      "uom":     "m/min",
      "current": 182,
      "target":  200
    },

    "energy": [
      {
        "name":         "Main Line",
        "streamId":     null,
        "jobRef":       "JOB-2026-4711",
        "articleRef":   "BC-200-140",
        "water":        { "value": 1.24,  "uom": "m³"     },
        "electricity":  { "value": 5200,  "uom": "Wh"     },
        "steam":        { "value": 340,   "uom": "kg"     },
        "air":          { "value": 0.92,  "uom": "m³"     },
        "stacks":       { "value": 48,    "uom": "stacks" }
      }
    ],

    "inputs": [
      {
        "name":           "Fluting Medium -- Roll 1",
        "streamId":       "IN-FM-1",
        "jobRef":         "JOB-2026-4711",
        "articleRef":     "BC-200-140",
        "area":           { "value": 19240, "uom": "m²" },
        "weight":         { "value": 3.85,  "uom": "t"  },
        "runningLength": { "value": 13032, "uom": "m"  }
      },
      {
        "name":           "Top Liner -- Roll 2",
        "streamId":       "IN-TL-2",
        "jobRef":         "JOB-2026-4711",
        "articleRef":     "BC-200-140",
        "area":           { "value": 19240, "uom": "m²" },
        "weight":         { "value": 4.04,  "uom": "t"  },
        "runningLength": { "value": 13032, "uom": "m"  }
      }
    ],

    "outputs": [
      {
        "name":           "Corrugated Board -- Stacker",
        "streamId":       "OUT-BD-1",
        "jobRef":         "JOB-2026-4711",
        "articleRef":     "BC-200-140",
        "area":           { "value": 18560, "uom": "m²"    },
        "weight":         { "value": 7.42,  "uom": "t"     },
        "runningLength": { "value": 12570, "uom": "m"     },
        "pieces":         { "value": 5024,  "uom": "pcs"   },
        "stacks":         { "value": 48,    "uom": "stacks"}
      }
    ],

    "logisticsStatus": [
      {
        "name":         "Transfer Belt -- Stacker to Storage",
        "streamId":     "LS-TB-01",
```

```
        "statusId":    "LS-01",
        "movement":    "running",
        "articleRef":  "BC-200-140",
        "area":        { "value": 920, "uom": "m²"     },
        "pieces":      { "value": 240, "uom": "pcs"    },
        "stacks":      { "value": 2,   "uom": "stacks" }
      }
    ]
  }
}
```

2.5 quality/quality.json

Source: examples/quality/quality.json


```
{
  "messageID":     "a9c12f3d-44b8-5e7a-bf2e-3c781d02e5f9",
  "correlationID": "7e4d2a18-99f3-5c1a-a3e0-1b6f7c8d5e2a",
  "assetClass":    "quality",
  "schemaId":      "cosi:quality-payload:1.0",
  "messageType":   "cos",
  "machineId":     "ffg-line-02",
  "timestamp":     "2026-05-04T07:45:00.000Z",
  "status":        "run",
  "payload": {
    "jobRef":     "JOB-2026-4711",
    "articleRef": "ART-0201-BC",
    "taskRef":    "TASK-FFG-001",
    "setupRef":   null,

    "qualityChecks": {

      "corrugating": {
        "warp": {
          "errorType": "Warp", "lowerLimit": null, "upperLimit": 8.0, "uom": "%",
          "checks": [
            { "quantity": 2422, "uom": "pcs", "result": "OK" }
          ]
        },
        "humidity": {
          "errorType": "Humidity", "lowerLimit": 45, "upperLimit": 60, "uom": "%RH",
          "checks": [
            { "quantity": 2422, "uom": "pcs", "result": "OK" }
          ]
        },
        "temperature": {
          "errorType": "Temperature", "lowerLimit": 18, "upperLimit": 28, "uom": "°C",
          "checks": [
            { "quantity": 2422, "uom": "pcs", "result": "OK" }
          ]
        },
        "caliper": {
          "errorType": "Caliper", "lowerLimit": 3.8, "upperLimit": 4.2, "uom": "mm",
          "checks": [
            { "quantity": 8,    "uom": "pcs", "result": "NOK", "ejected": true,  "exception":
            ↪  "lower-violation" },
            { "quantity": 14,   "uom": "pcs", "result": "NOK", "ejected": false, "exception":
            ↪  "upper-violation" },
            { "quantity": 2400, "uom": "pcs", "result": "OK" }
          ]
        },
        "paperQuality": {
          "errorType": "Paper quality", "lowerLimit": null, "upperLimit": null, "uom": null,
          "checks": [
            { "quantity": 2420, "uom": "pcs", "result": "OK" },
            { "quantity": 2, "uom": "pcs", "result": "NOK", "ejected": false }
          ]
        },
        "reelQuality": {
          "errorType": "Reel quality", "lowerLimit": null, "upperLimit": null, "uom": null,
          "checks": [
            { "quantity": 2422, "uom": "pcs", "result": "OK" }
          ]
        },
        "paperAlignment": {
          "errorType": "Paper alignment", "lowerLimit": -2.0, "upperLimit": 2.0, "uom": "mm",
          "checks": [
            { "quantity": 2422, "uom": "pcs", "result": "OK" }
          ]
        },
        "printRegister": {
          "errorType": "Print register", "lowerLimit": -1.5, "upperLimit": 1.5, "uom": "mm",
          "checks": [
            { "quantity": 2422, "uom": "pcs", "result": "OK" }
```

```
          ]
        },
        "blistersAndGluing": {
          "errorType": "Blisters and gluing", "lowerLimit": null, "upperLimit": null, "uom":
          ↪  null,
          "checks": [
            { "quantity": 2422, "uom": "pcs", "result": "OK" }
          ]
        },
        "cutQuality": {
          "errorType": "Cut quality", "lowerLimit": null, "upperLimit": null, "uom": null,
          "checks": [
            { "quantity": 2422, "uom": "pcs", "result": "OK" }
          ]
        },
        "scoring": {
          "errorType": "Scoring", "lowerLimit": -1.0, "upperLimit": 1.0, "uom": "mm",
          "checks": [
            { "quantity": 2422, "uom": "pcs", "result": "OK" }
          ]
        },
        "format": {
          "errorType": "Format", "lowerLimit": -2.0, "upperLimit": 2.0, "uom": "mm",
          "checks": [
            { "quantity": 2422, "uom": "pcs", "result": "OK" }
          ]
        }
      },

      "foldingGluing": {
        "gapSize": {
          "errorType": "Gap size", "lowerLimit": -1.0, "upperLimit": 1.0, "uom": "mm",
          "checks": [ { "quantity": 2422, "uom": "pcs", "result": "OK" } ]
        },
        "glueApplied": {
          "errorType": "Glue applied", "lowerLimit": null, "upperLimit": null, "uom": null,
          "checks": [ { "quantity": 2422, "uom": "pcs", "result": "OK" } ]
        },
        "scoring": {
          "errorType": "Scoring", "lowerLimit": -1.5, "upperLimit": 1.5, "uom": "mm",
          "checks": [ { "quantity": 2422, "uom": "pcs", "result": "OK" } ]
        },
        "fishtailing": {
          "errorType": "Fishtailing", "lowerLimit": null, "upperLimit": 3.0, "uom": "%",
          "checks": [ { "quantity": 2422, "uom": "pcs", "result": "OK" } ]
        },
        "wasteFree": {
          "errorType": "Waste free", "lowerLimit": 98.0, "upperLimit": null, "uom": "%",
          "checks": [ { "quantity": 2422, "uom": "pcs", "result": "OK" } ]
        }
      },

      "dieCutting": {
        "missingLinesCuttingCreasing": {
          "errorType": "Missing lines cutting/creasing", "lowerLimit": null, "upperLimit":
          ↪  null, "uom": null,
          "checks": [ { "quantity": 2422, "uom": "pcs", "result": "OK" } ]
        },
        "creasingCracking": {
          "errorType": "Creasing cracking", "lowerLimit": null, "upperLimit": null, "uom":
          ↪  null,
          "checks": [ { "quantity": 2422, "uom": "pcs", "result": "OK" } ]
        },
        "dimensions": {
          "errorType": "Dimensions", "lowerLimit": -2.0, "upperLimit": 2.0, "uom": "mm",
          "checks": [ { "quantity": 2422, "uom": "pcs", "result": "OK" } ]
        }
      },

      "logistics": {
        "stackAccuracy": {
          "errorType": "Stack accuracy", "lowerLimit": 95.0, "upperLimit": null, "uom": "%",
          "checks": [ { "quantity": 2422, "uom": "pcs", "result": "OK" } ]
        },
```

```
        "stackHeight": {
          "errorType": "Stack height", "lowerLimit": 80, "upperLimit": 200, "uom": "mm",
          "checks": [ { "quantity": 2422, "uom": "pcs", "result": "OK" } ]
        },
        "elephantFoot": {
          "errorType": "Elephant foot", "lowerLimit": null, "upperLimit": null, "uom": null,
          "checks": [ { "quantity": 2422, "uom": "pcs", "result": "OK" } ]
        },
        "labelsPosition": {
          "errorType": "Labels position", "lowerLimit": null, "upperLimit": null, "uom":
          ↪  null,
          "checks": [ { "quantity": 2422, "uom": "pcs", "result": "OK" } ]
        }
      },

      "printing": {
        "colors": [
          {
            "colorId": "station-1",
            "printingDefects": {
              "errorType": "Printing defects", "lowerLimit": null, "upperLimit": 2.0, "uom":
              ↪  null,
              "checks": [ { "quantity": 2422, "uom": "pcs", "result": "OK" } ]
            },
            "registerAccuracy": {
              "errorType": "Register accuracy", "lowerLimit": -1.0, "upperLimit": 1.0, "uom":
              ↪  "mm",
              "checks": [ { "quantity": 2422, "uom": "pcs", "result": "OK" } ]
            },
            "colorAccuracy": {
              "errorType": "Color accuracy", "lowerLimit": null, "upperLimit": 3.0, "uom":
              ↪  "dE2000",
              "checks": [ { "quantity": 2422, "uom": "pcs", "result": "OK" } ]
            },
            "inkTemperature": {
              "errorType": "Ink temperature", "lowerLimit": 18, "upperLimit": 26, "uom":
              ↪  "°C",
              "checks": [ { "quantity": 2422, "uom": "pcs", "result": "OK" } ]
            },
            "inkViscosity": {
              "errorType": "Ink viscosity", "lowerLimit": 18, "upperLimit": 24, "uom": "s",
              "checks": [ { "quantity": 2422, "uom": "pcs", "result": "OK" } ]
            },
            "inkPH": {
              "errorType": "Ink pH", "lowerLimit": 8.0, "upperLimit": 9.5, "uom": null,
              "checks": [ { "quantity": 2422, "uom": "pcs", "result": "OK" } ]
            }
          },
          {
            "colorId": "station-2",
            "printingDefects": {
              "errorType": "Printing defects", "lowerLimit": null, "upperLimit": 2.0, "uom":
              ↪  null,
              "checks": [ { "quantity": 2422, "uom": "pcs", "result": "OK" } ]
            },
            "registerAccuracy": {
              "errorType": "Register accuracy", "lowerLimit": -1.0, "upperLimit": 1.0, "uom":
              ↪  "mm",
              "checks": [ { "quantity": 2422, "uom": "pcs", "result": "OK" } ]
            },
            "colorAccuracy": {
              "errorType": "Color accuracy", "lowerLimit": null, "upperLimit": 3.0, "uom":
              ↪  "dE2000",
              "checks": [
                { "quantity": 22,   "uom": "pcs", "result": "NOK", "ejected": true,
                ↪  "exception": "upper-violation" },
                { "quantity": 2400, "uom": "pcs", "result": "OK" }
              ]
            },
            "inkTemperature": {
              "errorType": "Ink temperature", "lowerLimit": 18, "upperLimit": 26, "uom":
              ↪  "°C",
              "checks": [ { "quantity": 2422, "uom": "pcs", "result": "OK" } ]
            },
```

```
              "inkViscosity": {
                "errorType": "Ink viscosity", "lowerLimit": 18, "upperLimit": 24, "uom": "s",
                "checks": [ { "quantity": 2422, "uom": "pcs", "result": "OK" } ]
              },
              "inkPH": {
                "errorType": "Ink pH", "lowerLimit": 8.0, "upperLimit": 9.5, "uom": null,
                "checks": [ { "quantity": 2422, "uom": "pcs", "result": "OK" } ]
              }
            }
          ]
        }

    }
  }
}
```

2.6 setup/setup-converting.json

Source: examples/setup/setup-converting.json

```
{
  "messageID":     "d4e5f6a7-b8c9-5d1e-af2b-3c4d5e6f7a8b",
  "assetClass":    "setup-converting",
  "schemaId":      "cosi:setup-converting-payload:1.0",
  "messageType":   "cos",
  "machineId":     "ffg-line-02",
  "timestamp":     "2026-05-04T06:30:00.000Z",
  "status":        "setup",
  "payload": {

    "customProps": {},

    "quickSetId": null,

    "jobSettings": {
      "jobRef":     "JOB-2026-4711",
      "orderRef":   "ORD-2026-4711",
      "customer":   "Acme Packaging GmbH",
      "dueDate":    "2026-05-04T14:00:00.000Z",
      "quantity": {
        "uom":       "pcs",
        "scheduled": 5000
      },
      "autoStopFeed": true
    },

    "articleSettings": {
      "articleRef":  "ART-0201-BC",
      "boxType":     { "value": 201,  "uom": "FEFCO" },
      "boxWidth":    { "value": 300,  "uom": "mm" },
      "boxLength":   { "value": 400,  "uom": "mm" },
      "boxHeight":   { "value": 200,  "uom": "mm" },
      "glueTab":     { "value": 36,   "uom": "mm" },
      "register":    { "value": 0,    "uom": "mm" },
      "panels": [
        { "panelId": "A",  "value": 400, "uom": "mm" },
        { "panelId": "B",  "value": 300, "uom": "mm" },
        { "panelId": "C",  "value": 400, "uom": "mm" },
        { "panelId": "D",  "value": 300, "uom": "mm" },
        { "panelId": "A1", "value": 36,  "uom": "mm" }
      ]
    },

    "rawMaterial": {
      "flute":          "BC",
      "grade":          "KLB 200 / SC 140 / KLB 200",
      "boardThickness": { "value": 5.6,  "uom": "mm" },
      "surfaceType":    null,
      "crossFlute":     false,
      "papers": [
        { "layer": "liner-top",    "gsm": { "value": 200, "uom": "g/m²" }, "paperType":
        ↪  "Kraftliner Brown" },
        { "layer": "fluting",      "gsm": { "value": 140, "uom": "g/m²" }, "paperType":
        ↪  "Semichemical"     },
        { "layer": "liner-bottom", "gsm": { "value": 200, "uom": "g/m²" }, "paperType":
        ↪  "Kraftliner Brown" }
      ]
    },

    "machineSettings": {
      "taskRef":      "TASK-FFG-001",
      "stageNo":      1,
      "outputWidth":  { "value": 736,  "uom": "mm" },
      "outputLength": { "value": 600,  "uom": "mm" },
      "infeedSettings": {
        "active":      true,
        "feedType":    "vacuum",
        "inputWidth":  { "value": 1480, "uom": "mm" },
        "inputLength": { "value": 1200, "uom": "mm" },
```

```
        "numberIn":    1
      },
      "speed": {
        "uom":       "pcs/h",
        "target":    9000,
        "scheduled": 8500
      },
      "setupDuration": {
        "uom":       "s",
        "target":    300,
        "scheduled": 420
      }
    },

    "periphery": [
      {
        "deviceId":   "print-1",
        "deviceType": "flexo-print",
        "settings": {
          "active":              true,
          "color":               { "value": "FF1B1B1B", "uom": "ARGB" },
          "inkName":             "Process Black",
          "plateId":             "PLT-4711-BLK",
          "position":            { "value": 1, "uom": "station" },
          "register":            { "value": 0.5, "uom": "mm" },
          "cylinderCentering":   { "value": 0,   "uom": "mm" },
          "cylinderCalibration": { "value": 0.2, "uom": "mm" },
          "aniloxCalibration":   { "value": 0,   "uom": "mm" },
          "useDryer":            true,
          "plateHeight":         { "value": 3.18, "uom": "mm" }
        }
      },
      {
        "deviceId":   "print-2",
        "deviceType": "flexo-print",
        "settings": {
          "active":              true,
          "color":               { "value": "FFCC0000", "uom": "ARGB" },
          "inkName":             "Pantone 485 Red",
          "plateId":             "PLT-4711-RED",
          "position":            { "value": 2, "uom": "station" },
          "register":            { "value": 0.3, "uom": "mm" },
          "cylinderCentering":   { "value": 0,   "uom": "mm" },
          "cylinderCalibration": { "value": 0.2, "uom": "mm" },
          "aniloxCalibration":   { "value": 0,   "uom": "mm" },
          "useDryer":            true,
          "plateHeight":         { "value": 3.18, "uom": "mm" }
        }
      },
      {
        "deviceId":   "digiprint",
        "deviceType": "digital-inkjet",
        "settings": {
          "active":          false,
          "printFileUrl":    "",
          "printWidth":      { "value": 0, "uom": "mm" },
          "printHeight":     { "value": 0, "uom": "mm" },
          "numberOfCopies":  0,
          "colorCorrection": null,
          "resolution":      { "value": 0, "uom": "dpi" },
          "substrate":       null,
          "orientation":     "TOP",
          "leadIn":          { "value": 0, "uom": "mm" },
          "leadOut":         { "value": 0, "uom": "mm" }
        }
      },
      {
        "deviceId":   "diecut",
        "deviceType": "die-cutter",
        "settings": {
          "active":        true,
          "cutterType":    "rotary",
          "upsWidth":      2,
          "upsLength":     1,
```

```
          "toolRef":       "DIE-4711-0201",
          "diePressure":   { "value": 14.5, "uom": "kN" },
          "cuttingForce":  { "value": 12.0, "uom": "kN" },
          "offsetScoring": { "value": 0.5,  "uom": "mm" },
          "scores": [
            { "scoreId": 1, "scoreType": "V-cut",  "position": { "value": 115,  "uom": "mm" } },
            { "scoreId": 2, "scoreType": "V-cut",  "position": { "value": 330,  "uom": "mm" } },
            { "scoreId": 3, "scoreType": "V-cut",  "position": { "value": 540,  "uom": "mm" } },
            { "scoreId": 4, "scoreType": "V-cut",  "position": { "value": 695,  "uom": "mm" }
          ]
        }
      },
      {
        "deviceId":   "folder-gluer",
        "deviceType": "folder-gluer",
        "settings": {
          "active":     true,
          "glueType":   1,
          "glueStyle":  "GI",
          "glueAmount": { "value": 4.5, "uom": "g/m²" }
        }
      },
      {
        "deviceId":   "stitcher",
        "deviceType": "stitcher",
        "settings": {
          "active":     false,
          "stitchType": 0
        }
      },
      {
        "deviceId":   "taper",
        "deviceType": "taper",
        "settings": {
          "active":   false,
          "tapeType": ""
        }
      },
      {
        "deviceId":   "bundlestrap",
        "deviceType": "bundle-strapper",
        "settings": {
          "active":                true,
          "itemsPerBundle":        { "value": 25,  "uom": "pcs" },
          "bundleWidth":           { "value": 750, "uom": "mm"  },
          "bundleHeight":          { "value": 380, "uom": "mm"  },
          "bundlePositionLateral": { "value": 375, "uom": "mm"  }
        }
      },
      {
        "deviceId":   "logistics",
        "deviceType": "palletizer",
        "settings": {
          "active":          true,
          "palletType":      1,
          "palletLength":    { "value": 1200, "uom": "mm" },
          "palletWidth":     { "value": 800,  "uom": "mm" },
          "maxHeight":       { "value": 1400, "uom": "mm" },
          "itemsPerPallet":  { "value": 200,  "uom": "pcs" },
          "numberOfLayers":  8,
          "packetsPerLayer": 4,
          "compositionRef":  "COMP-0201-EUR-200",
          "interleaves":     false,
          "strapping": {
            "active":         true,
            "positions":      [400, 800],
            "numberOfStraps": 2,
            "pressure":       { "value": 2.5, "uom": "kN" },
            "programId":      null
          },
```

```
          "wrapping": {
            "active":    true,
            "programId": "WRAP-STD-01",
            "wrapType":  1
          },
          "counterPallet":    false,
          "coverSheet":       true,
          "bottomSheet":      true,
          "cornerProtection": false,
          "label": {
            "position": "OS-front",
            "data":     "JOB-2026-4711"
          }
        }
      }
    ]

  }
}
```

2.7 setup/setup-corrugator.json

Source: examples/setup/setup-corrugator.json


```
{
  "messageID":     "7c8b2a1e-3d4f-5e5c-8b9a-1f2e3d4c5b6a",
  "assetClass":    "setup-corrugator",
  "schemaId":      "cosi:setup-corrugator-payload:1.0",
  "messageType":   "cos",
  "machineId":     "corr-line-01",
  "timestamp":     "2026-05-04T06:12:00.000Z",
  "status":        "setup",
  "payload": {

    "customProps": {},

    "setupSettings": {
      "setupId":        "SETUP-2026-0881",

      "webWidth":   { "value": 1600, "uom": "mm" },
      "leftTrim":   { "value": 10,   "uom": "mm" },
      "rightTrim":  { "value": 10,   "uom": "mm" },

      "speed": {
        "uom":     "m/min",
        "target":  200
      },
      "control1Speed": { "value": 178, "uom": "m/min" },
      "control2Speed": { "value": 174, "uom": "m/min" },

      "runningLength": {
        "uom":       "m",
        "scheduled": 14400
      }
    },

    "boardGradeSettings": {
      "boardGradeRef": "BC-200-140",
      "fluteType":     "BC",
      "caliper":       { "value": 5.6, "uom": "mm" }
    },

    "orders": [
      {
        "customProps": {},

        "orderType":         "production",
        "orderRef":          "ORD-2026-4711",
        "partRef":           "PART-001",
        "articleRef":        "ART-0201-BC",
        "customerName":      "Acme Packaging GmbH",
        "processingMachine": "ffg-line-02",
        "destinationLine":   2,
        "l2rSequenceNumber": 1,

        "sheetWidth":  { "value": 1480, "uom": "mm" },
        "sheetLength": { "value": 1200, "uom": "mm" },

        "cuts": {
          "uom":    "cuts",
          "target": 5200
        },

        "numberOfOuts":    2,
        "cutoffAllocation": 0,

        "preprint": {
          "active":         false,
          "markDefinition": 0,
          "mark1Width":  { "value": 0, "uom": "mm" },
          "gap1Width":   { "value": 0, "uom": "mm" },
          "mark2Width":  { "value": 0, "uom": "mm" },
          "gap2Width":   { "value": 0, "uom": "mm" },
```

```
          "mark3Width":  { "value": 0, "uom": "mm" },
          "windowWidth": { "value": 0, "uom": "mm" },
          "markToEdge":  { "value": 0, "uom": "mm" },
          "markOffset":  { "value": 0, "uom": "mm" },
          "printSide":   "bottom",
          "colorContrast": 0
        },

        "scoringProfiles": [
          { "profileId": 1, "profileType": "V-cut", "offset": 0, "depth": { "value": 1.2, "uom": "mm" } },
          { "profileId": 2, "profileType": null,    "offset": 0, "depth": { "value": 0, "uom": "mm" } }
        ],

        "scoringPositions": [
          { "position":  1, "value": 115,  "uom": "mm", "profileId": 1 },
          { "position":  2, "value": 330,  "uom": "mm", "profileId": 1 },
          { "position":  3, "value": 540,  "uom": "mm", "profileId": 1 },
          { "position":  4, "value": 755,  "uom": "mm", "profileId": 1 },
          { "position":  5, "value": 855,  "uom": "mm", "profileId": 1 },
          { "position":  6, "value": 1070, "uom": "mm", "profileId": 1 },
          { "position":  7, "value": 1280, "uom": "mm", "profileId": 1 },
          { "position":  8, "value": 1395, "uom": "mm", "profileId": 1 },
          { "position":  9, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 10, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 11, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 12, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 13, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 14, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 15, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 16, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 17, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 18, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 19, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 20, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 21, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 22, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 23, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 24, "value": 0,    "uom": "mm", "profileId": 1 }
        ],

        "stackHeight":        { "value": 800, "uom": "mm"    },
        "stackCount":         { "value": 200, "uom": "pcs" },
        "lastRun":            false,
        "dischargeDirection": "OS",
        "intermediateLayer":  false,
        "palletType":         "Euro"
      },
      {
        "orderType":         "production",
        "orderRef":          "ORD-2026-4712",
        "partRef":           "PART-001",
        "articleRef":        "ART-0201-BC-S",
        "customerName":      "Beta Logistics AG",
        "processingMachine": "ffg-line-02",
        "destinationLine":   2,
        "l2rSequenceNumber": 2,

        "sheetWidth":  { "value": 1480, "uom": "mm" },
        "sheetLength": { "value": 960,  "uom": "mm" },

        "cuts": {
          "uom":    "cuts",
          "target": 3800
        },

        "numberOfOuts":    2,
        "cutoffAllocation": 0,

        "preprint": {
          "active":         false,
          "markDefinition": 0,
          "mark1Width":  { "value": 0, "uom": "mm" },
```

```
          "gap1Width":   { "value": 0, "uom": "mm" },
          "mark2Width":  { "value": 0, "uom": "mm" },
          "gap2Width":   { "value": 0, "uom": "mm" },
          "mark3Width":  { "value": 0, "uom": "mm" },
          "windowWidth": { "value": 0, "uom": "mm" },
          "markToEdge":  { "value": 0, "uom": "mm" },
          "markOffset":  { "value": 0, "uom": "mm" },
          "printSide":   "bottom",
          "colorContrast": 0
        },

        "scoringProfiles": [
          { "profileId": 1, "profileType": "V-cut", "offset": 0,  "depth": { "value": 1.2,
          ↪  "uom": "mm" } },
          { "profileId": 2, "profileType": null,    "offset": 0,  "depth": { "value": 0,
          ↪  "uom": "mm" } }
        ],

        "scoringPositions": [
          { "position":  1, "value": 95,   "uom": "mm", "profileId": 1 },
          { "position":  2, "value": 280,  "uom": "mm", "profileId": 1 },
          { "position":  3, "value": 440,  "uom": "mm", "profileId": 1 },
          { "position":  4, "value": 625,  "uom": "mm", "profileId": 1 },
          { "position":  5, "value": 720,  "uom": "mm", "profileId": 1 },
          { "position":  6, "value": 905,  "uom": "mm", "profileId": 1 },
          { "position":  7, "value": 1065, "uom": "mm", "profileId": 1 },
          { "position":  8, "value": 1250, "uom": "mm", "profileId": 1 },
          { "position":  9, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 10, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 11, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 12, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 13, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 14, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 15, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 16, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 17, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 18, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 19, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 20, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 21, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 22, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 23, "value": 0,    "uom": "mm", "profileId": 1 },
          { "position": 24, "value": 0,    "uom": "mm", "profileId": 1 }
        ],

        "stackHeight":        { "value": 600, "uom": "mm"    },
        "stackCount":         { "value": 150, "uom": "pcs" },
        "lastRun":            true,
        "dischargeDirection": "DS",
        "intermediateLayer":  false,
        "palletType":         "Euro"
      }
    ],

    "periphery": [
      {
        "deviceId":   "SF1",
        "deviceType": "single-facer",
        "settings": {
          "active": true,
          "linerPaper": {
            "paperId":     "KLB-200-SF1",
            "gsm":         { "value": 200, "uom": "g/m²" },
            "caliper":     { "value": 0.31, "uom": "mm" },
            "width":       { "value": 1620, "uom": "mm" },
            "requiredFSC": "100%"
          },
          "mediumPaper": {
            "paperId":     "SC-140-SF1",
            "gsm":         { "value": 140, "uom": "g/m²" },
            "caliper":     { "value": 0.22, "uom": "mm" },
            "width":       { "value": 1620, "uom": "mm" },
            "requiredFSC": null
          },
```

```
          "fluteType": "B",
          "processAdds": {
            "waterResistant": false,
            "whiteTop":       false,
            "wax":            false,
            "coating":        false
          }
        }
      },
      {
        "deviceId":   "SF2",
        "deviceType": "single-facer",
        "settings": {
          "active": false,
          "linerPaper": {
            "paperId":     null,
            "gsm":         { "value": 0, "uom": "g/m²" },
            "caliper":     { "value": 0, "uom": "mm" },
            "width":       { "value": 0, "uom": "mm" },
            "requiredFSC": null
          },
          "mediumPaper": {
            "paperId":     null,
            "gsm":         { "value": 0, "uom": "g/m²" },
            "caliper":     { "value": 0, "uom": "mm" },
            "width":       { "value": 0, "uom": "mm" },
            "requiredFSC": null
          },
          "fluteType": null,
          "processAdds": {
            "waterResistant": false,
            "whiteTop":       false,
            "wax":            false,
            "coating":        false
          }
        }
      },
      {
        "deviceId":   "DB",
        "deviceType": "double-backer",
        "settings": {
          "active": true,
          "linerPaper": {
            "paperId":     "KLB-200-DB",
            "gsm":         { "value": 200, "uom": "g/m²" },
            "caliper":     { "value": 0.31, "uom": "mm" },
            "width":       { "value": 1620, "uom": "mm" },
            "requiredFSC": "100%"
          },
          "processAdds": {
            "waterResistant": false,
            "whiteTop":       false,
            "wax":            false,
            "coating":        false
          }
        }
      },
      {
        "deviceId":   "fanfold",
        "deviceType": "fanfold",
        "settings": {
          "active":        false,
          "orderIndex":    0,
          "numberOfFolds": 0,
          "creaseLength":  { "value": 0, "uom": "mm" }
        }
      },
      {
        "deviceId":   "tape",
        "deviceType": "tape",
        "settings": {
          "active":     false,
          "orderIndex": 0,
          "tapeType":   "",
```

```
            "positions":  []
          }
        }
      ]

    }
}
```

2.8 setup-response/setup-response-error.json

Source: examples/setup-response/setup-response-error.json


```
{
  "messageID":     "aa92d4b7-6f3c-5f2d-8fb2-5a11758ad531",
  "correlationID": "7c8b2a1e-3d4f-5e5c-8b9a-1f2e3d4c5b6a",
  "assetClass":    "setup-response",
  "schemaId":      "cosi:setup-response-payload:1.0",
  "messageType":   "cos",
  "machineId":     "corr-line-01",
  "timestamp":     "2026-07-17T10:16:00.000Z",
  "status":        "setup",
  "payload": {
    "result": "NOK",
    "errors": [
      {
        "fieldRef": "/payload/orders/0/sheetWidth/value",
        "description": "Width exceeds allowed range for current machine profile.",
        "manufacturerCode": 10042
      },
      {
        "fieldRef": "/payload/setupSettings/speed/target",
        "description": "Target speed is exceeding machine design limits.",
        "manufacturerCode": 10411
      }
    ]
  }
}
```

2.9 setup-response/setup-response.json

Source: examples/setup-response/setup-response.json

```
{
  "messageID":     "9c6d8f80-3b3b-5331-8f2a-4e6d70a5b123",
  "correlationID": "7c8b2a1e-3d4f-5e5c-8b9a-1f2e3d4c5b6a",
  "assetClass":    "setup-response",
  "schemaId":      "cosi:setup-response-payload:1.0",
  "messageType":   "cos",
  "machineId":     "corr-line-01",
  "timestamp":     "2026-07-17T10:15:00.000Z",
  "status":        "setup",
  "payload": {
    "result": "OK"
  }
}
```